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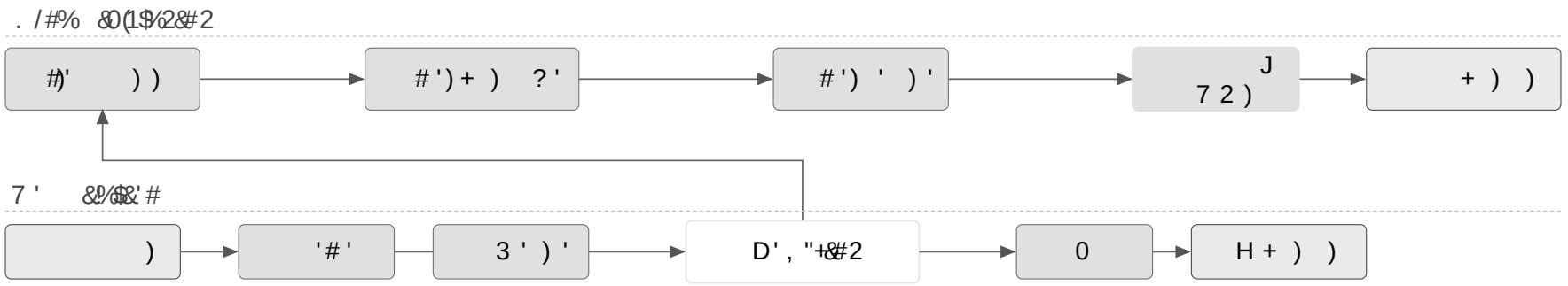

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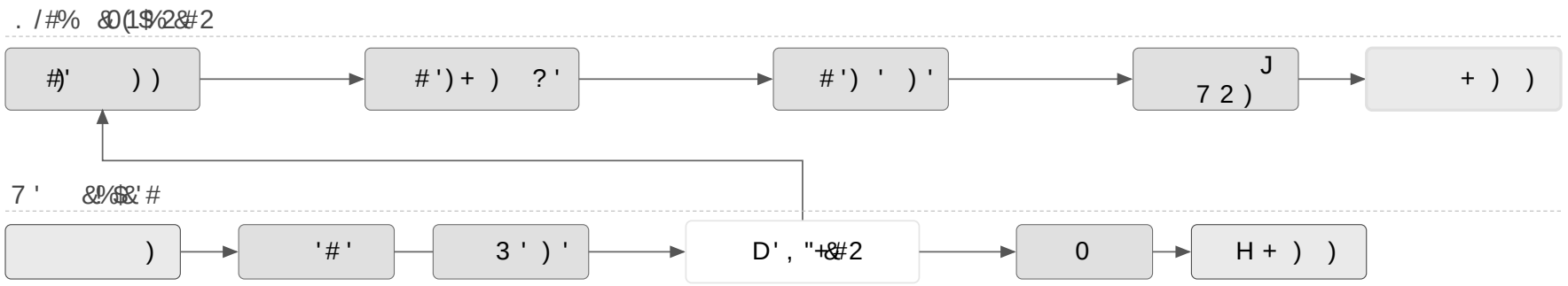
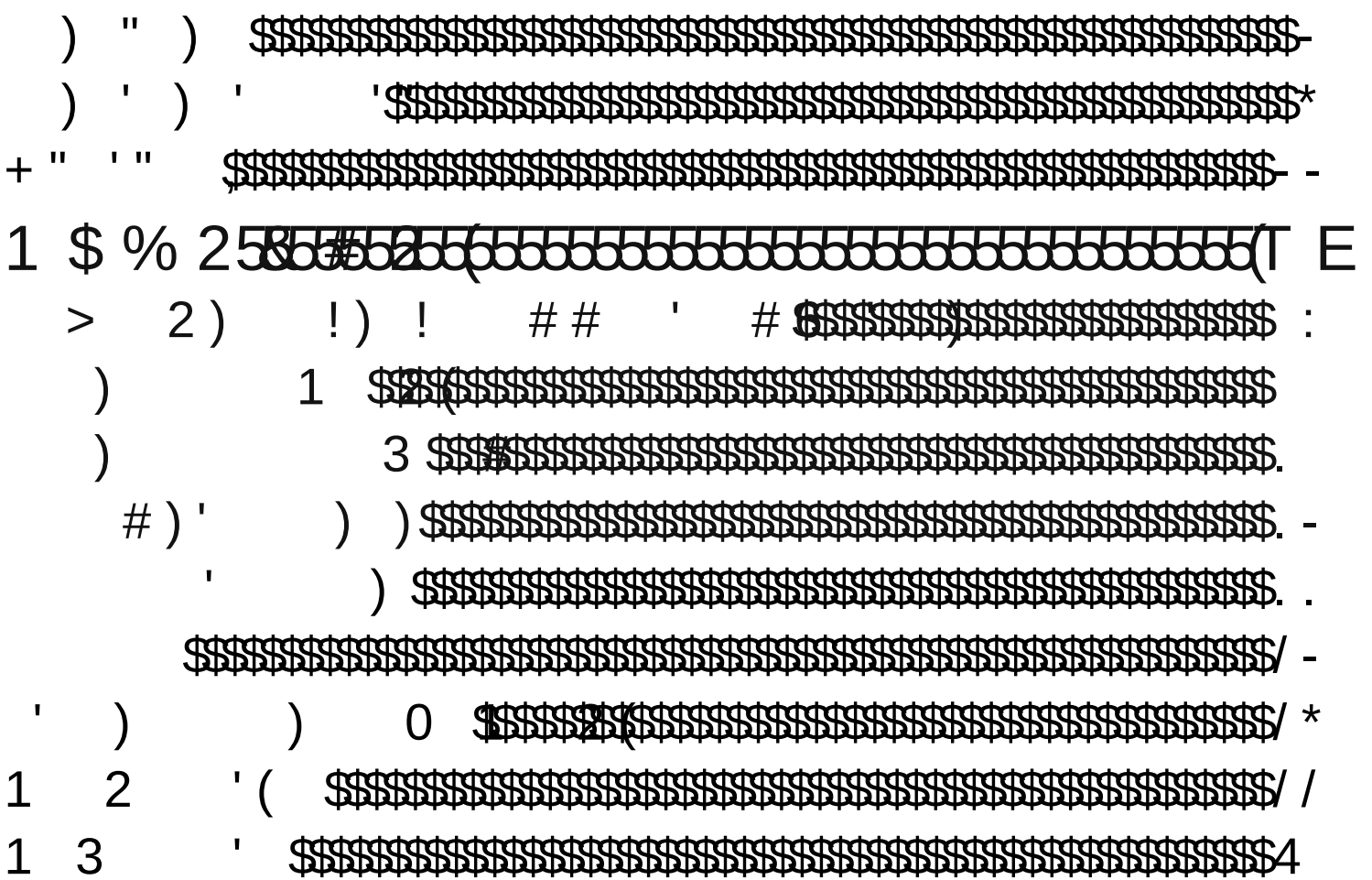


1 ' < + 0 " (3 D : 0 + & $ " # " + % $ ") (3 D : 0 + & $

```
if C(0) is                    mls
  Int  then "Int"
  C(n) then "C(" + n + ")"
  else "Unknown"
```

```
"C(0)"                        mls
```



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6 3 0 6 2) # 3) ')) 2 # # #

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```scala
case class Return(res, implct) extends Block
case class Match(scrut, arms, dflt, rest) extends Block
case class Assign(lhs, rhs, rest) extends Block
```

```mls
class Block with
  constructor
    Return(res, implct)
    Match(scrut, arms, dflt, rest)
    Assign(lhs, rhs, rest)
```

> ' 2 1 2 (0) , 2 ' 2')) # #)' 2) '
, 2 , 2 1 $0% 2 ") $(O ! ' 0 >

1 $ % 2 & # 2 (O ! ' 0 >

```
fun pow(x, n) = if n is 0 then 1 else x * pow(x, n-1)
```

```Scala
1  val pow = Symbol("pow"); val x = Symbol("x"); val n = Symbol("n")
2  val t1 = Symbol("tmp"); val t2 = Symbol("tmp")
3  val sub = Symbol("-"); val mul = Symbol("*")
4  FunDefn(pow, [x, n], Scoped(HashSet(t1, t2), Match(Ref(n),
5    [[ Lit(0), Return(Lit(1)) ]],
6    Assign(t1, Call(sub, [n, Lit(1)]),
7      Assign(t2, Call(pow, [x, t1]),
8        Return(Call(mul, [x, t2]))
9      )
10   ), End())
11 ))
```

G # # , ") 2 '' # 0 2 #)' 2) ' , 2 2 #)' 2) '))) 0 1 2 ($

1 $ % 2 & # 2 (O ! ' 0 >

```
fun pow(x, n) = if n is 0 then 1 else x * pow(x, n-1)
```

```
1   let pow = Symbol("pow"); let x = Symbol("x"); let n = Symbol("n")          mls
2   let t1 = Symbol("tmp"); let t2 = Symbol("tmp")
3   let sub = Symbol("-"); let mul = Symbol("*")
4   FunDefn(pow, [x, n], Scoped([t1, t2], Match(Ref(n),
5     [[ Lit(0), Return(Lit(1)) ]],
6     Assign(t1, Call(sub, [n, Lit(1)]),
7       Assign(t2, Call(pow, [x, t1]),
8         Return(Call(mul, [x, t2]))
9       )
10    ), End())
11  ))
```

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1 $ % 2 & # 2 (O ! ' 0 >

& '6 ' #)' 2) ' 0 2)) #) 2) 6 2

`Value.Lit(lit)`

```
ValueLit(lit)                    mls
```

7 1A `ValueLit(1)`

`Symbol(name)`

```
Symbol(name)                     mls
```

7 A

I x `Symbol("x")`

I c K L

) 6 ' #)

3 #$ & M ' " #)) # 2

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1 $ % 2 & # 2 (O ! ' 0 >

& '6 ' #)' 2) ' 0 2)) #) 2) 6 2

`Value.Lit(lit)`

```
ValueLit(lit)                                    mls
```

7 1A `ValueLit(1)`

`Value.Ref(sym)`

```
let l = sym                                      mls
ValueRef(l)
```

7 xA `ValueRef(Symbol("x"))`

`Select(qual, name)`

```
let q = qual                                     mls
let n = name
Select(q, n)
```

7 xAp `Select(ValueRef(Symbol("x")), Symbol("p"))`

1 $ % 2 & # 2 (O ! ' 0 >

```
Tuple([x0, x1, ...])
```

```
let s0 = x0                                                    mls
let s1 = x1
                                    …
Tuple([x0, x1, ...])
```

```
    ) ' 2   1 2( 2 # # ' #    ' #)     )    '  )
1 2( 3   0 2)     0 ' 2' )   )  2 '' #  0    )  0
2  ) '  ')$
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1 $ % 2 & # 2 (1 / A ' ! :

I & 0 ' 6 ') 3)) 3) ' #

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C.x

```mls
let CSym = ClassSymbol("C", C)
Select(ValueRef(CSym), Symbol("x"))
```

G 0 0 ' 0 ' 2) ,) 2 '') #)) , 2 2 ## 6

```mls
staged class A with
  fun f() = M.f()
  val redirect_M = M
```

1 $ % 2 & # 2 (1 / A ' ! :

I @ ## 6 3 #

& ,) 3 # 6 ') # 3 N 2)) ' " # #)

) 2 '') #) 2' ## 6 2) # 0 2) #$

> ' 2 # 3 # , 2) 0 ' #)) '

3 $

x + x

```mls
let x = Symbol("x")
// let x1 = Symbol("x")
Call(ValueRef(Symbol("+")), [[ValueRef(x), ValueRef(x)]])
```

1 $ % 2 & # 2 (1 / A ' ! :

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& ,) 3 # 6 ') # 3 N 2)) ' " # #)
) 2 '') #) 2' ## 6 2) # 0 2) #$

> ' 2 ## 0 0 # 3 # , 0) 2 2 0 # B'
3 ,)) #) 0 2 0 $

M.f()

```
1  let MSym = symbolMap.check(ModuleSymbol("M", M))
2  Call(Select(ValueRef(MSym), Symbol("f")), [[]])
```

mls

: $ + < " # $ % $ & '

') # 7 ' ' " ' 3 #96 2) #)) 0
') $

```
staged module Math with                                    mls
  val funCache = new Map()
  val generatorMap = new Map([["pow", pow_gen]])
  fun pow_staged() = ...
  fun pow_gen(x, n) =
    specialize(funCache, "f", pow_staged, [x, n])
  fun propagate() =
    let dyn = ...
    pow_gen(dyn, dyn)
    sq_gen(dyn)
```

: $ + < " # $ % $ & '

') # 7 ' ' " ' 3 #96 2) #)) 0
') $

?) # 3 #*$0

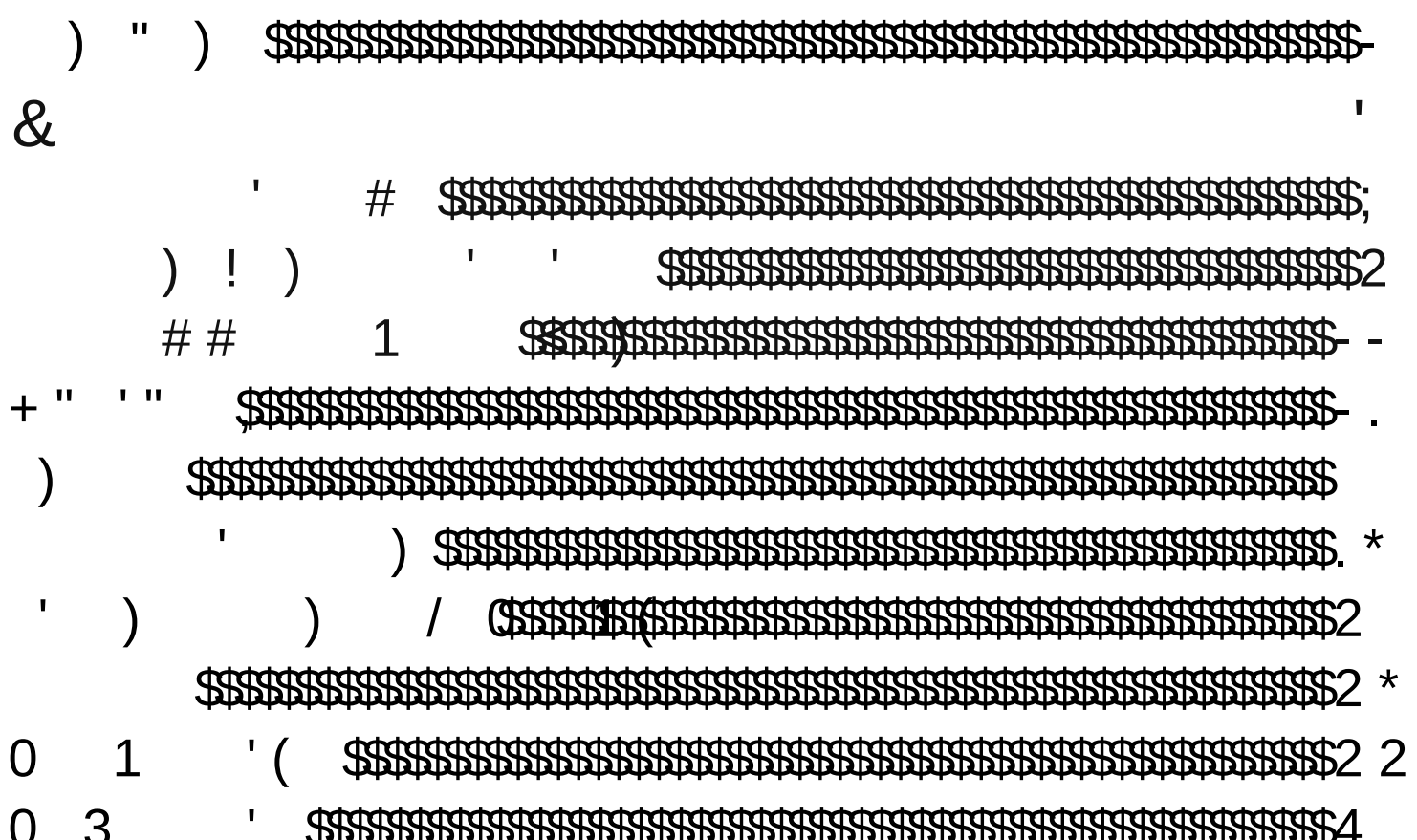
```mls
staged module Math with
  val funCache // memoize specialized functions
  val generatorMap // point to generator function
  fun pow_staged() // return staged version of the function
  fun pow_gen(x, n) // redirect to shape propagation
  fun propagate() // generates all entry functions
```

: $ + < " # $ % $ & '

> ' #) 0 2 ## # , 2)')) ') '# 6) 2 #
') ') # 2 ?) 6 2) # # # $

```mls
staged class C(x, y) with
  fun add() = x + y
```

```mls
staged module C with
  fun add(cls)() = cls.x + cls.y
```



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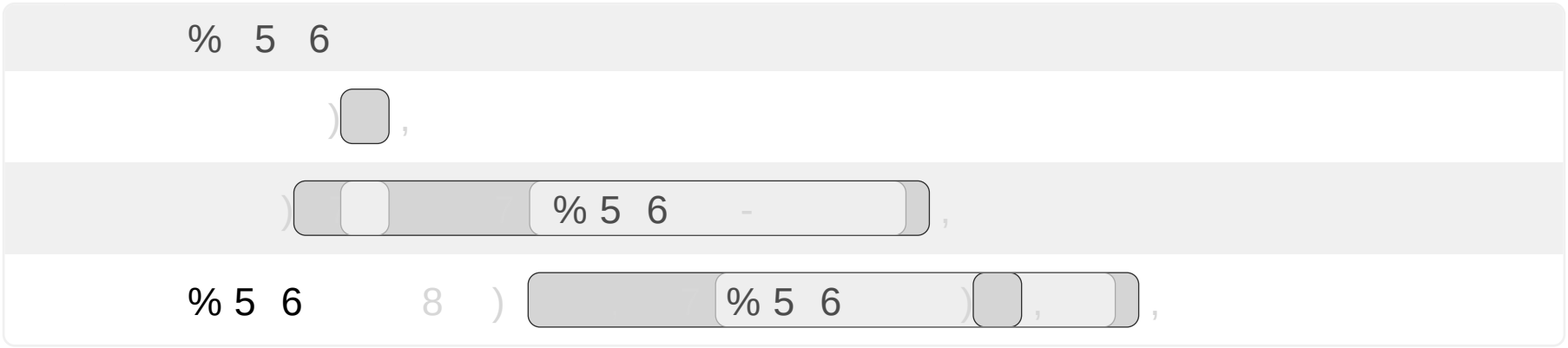

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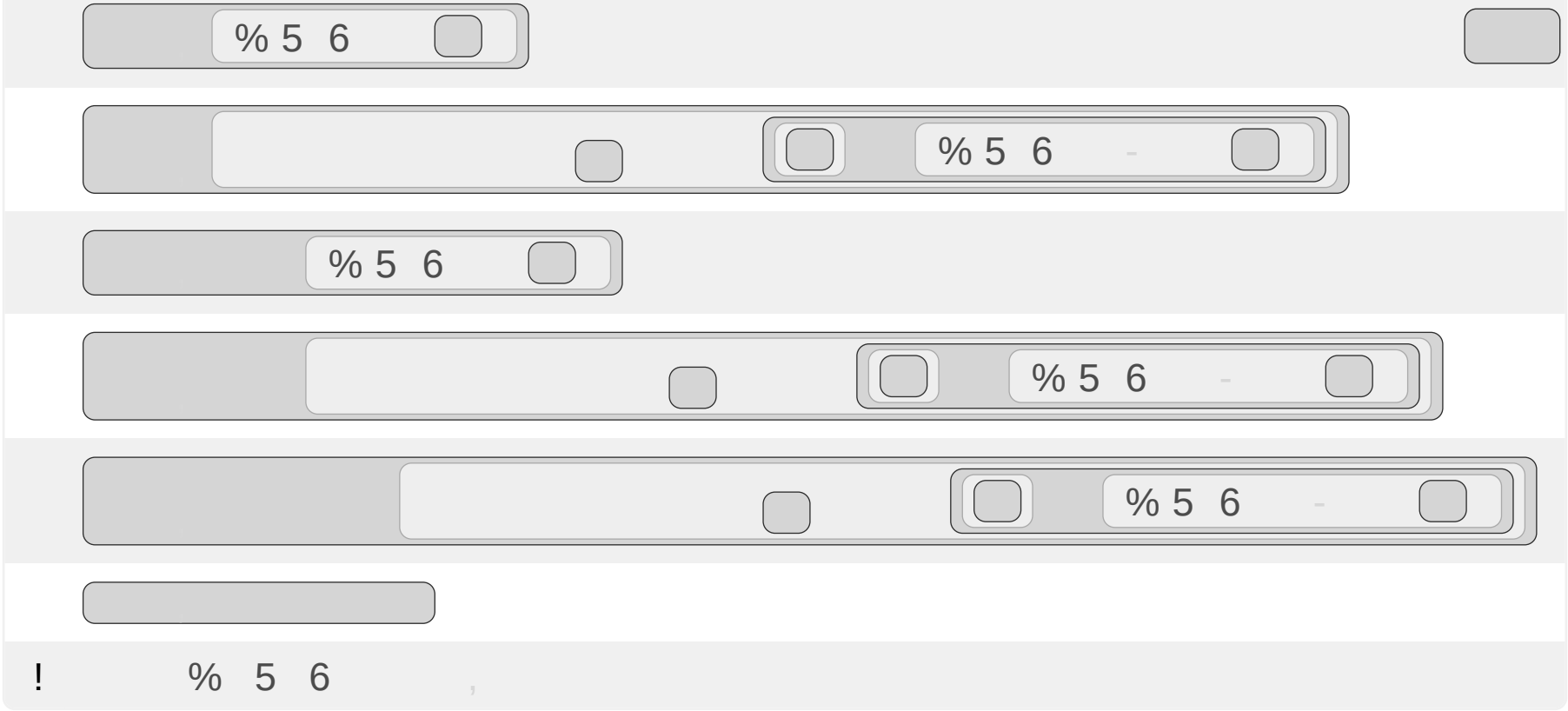

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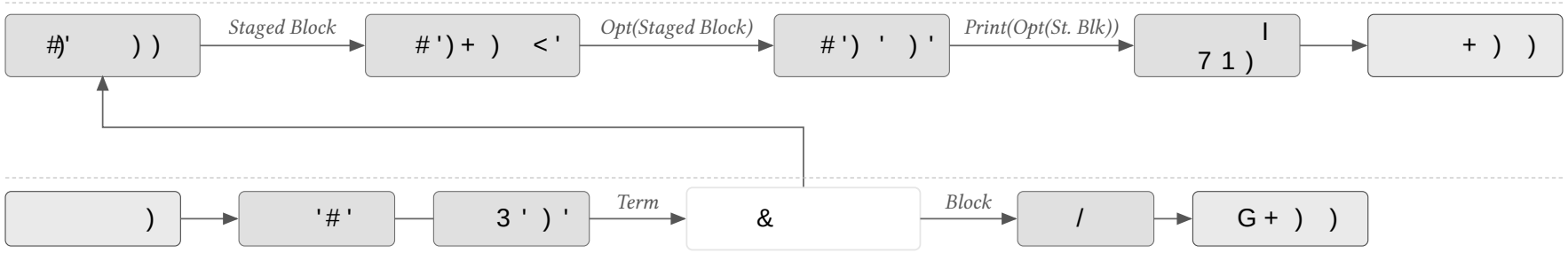
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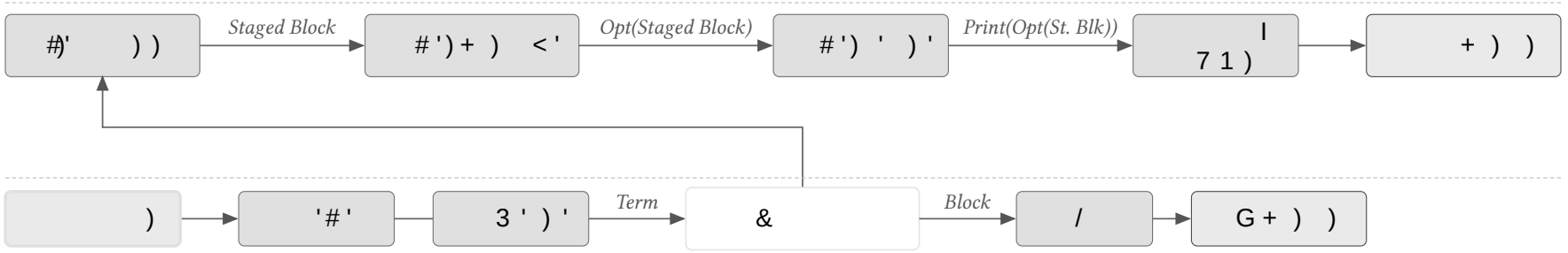

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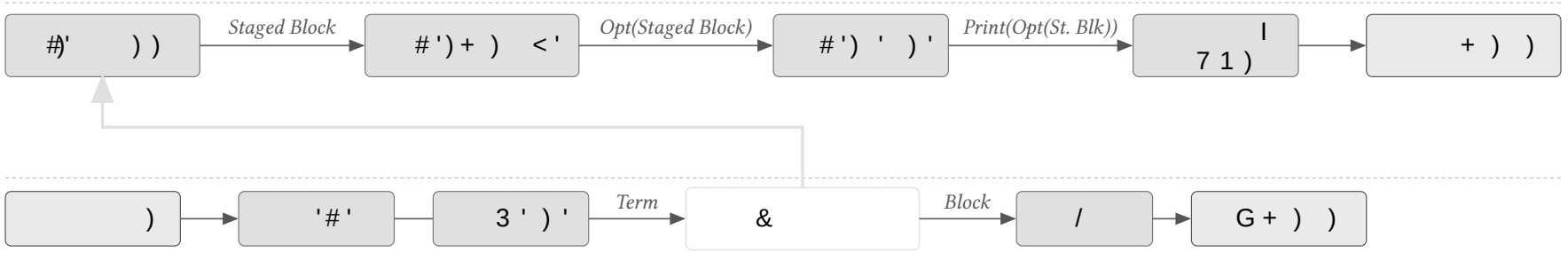

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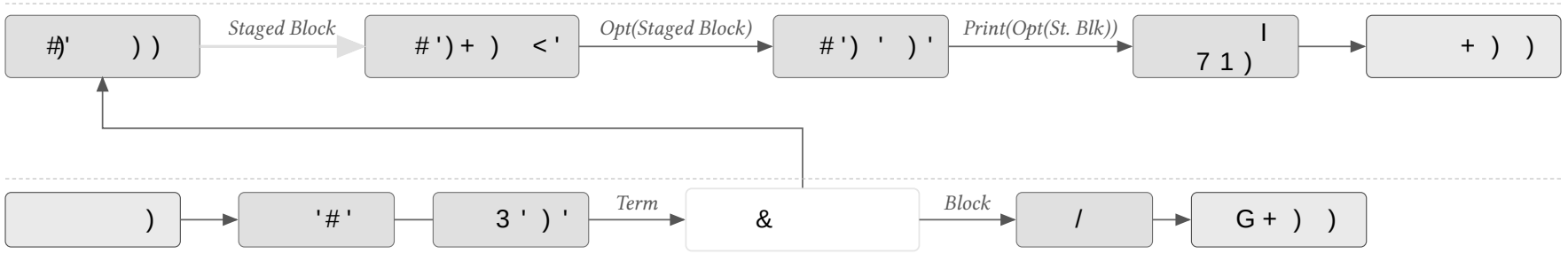

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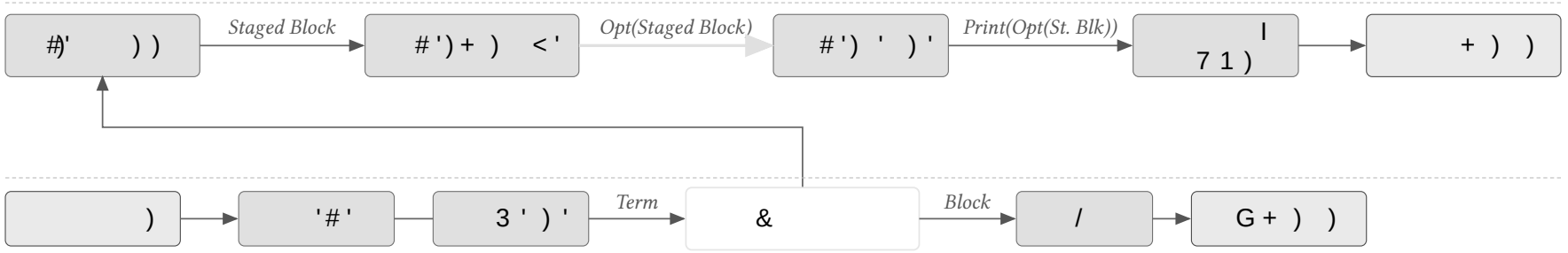
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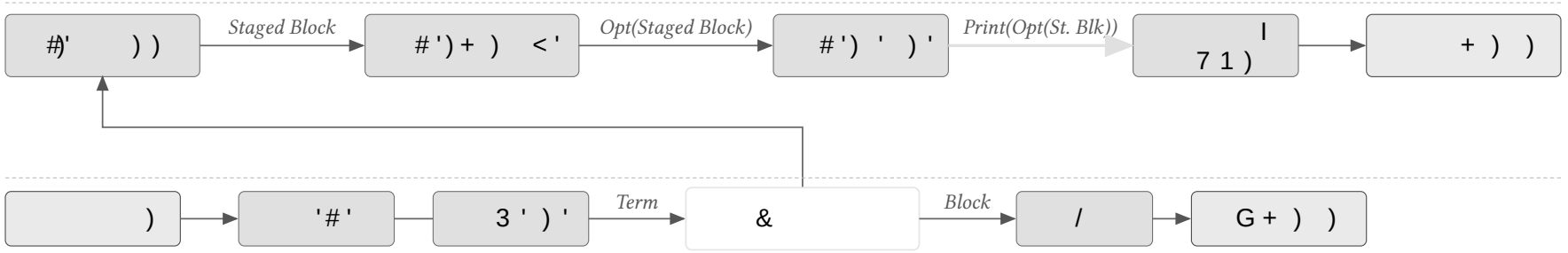
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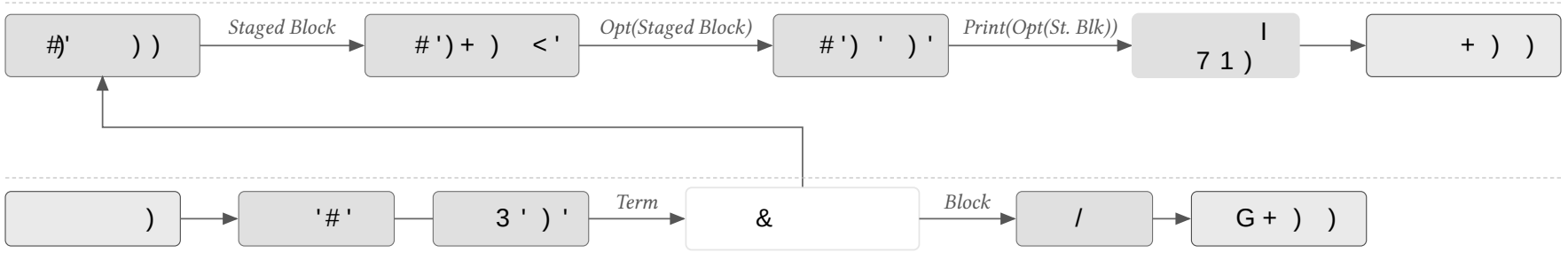

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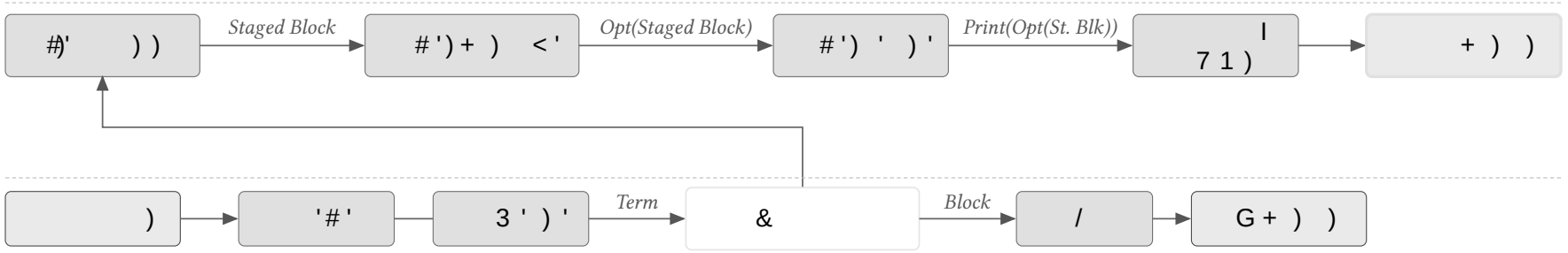

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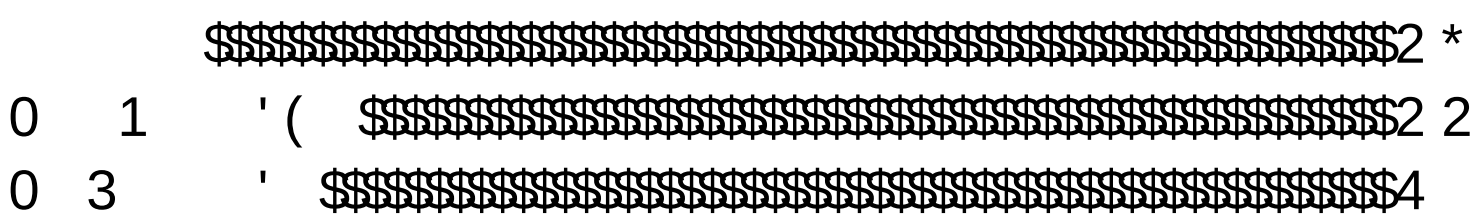

- ! - 8 6

Symbol '()(*

Literal +

Function definition ,- . /0+−1 2 3 4 5

Plain class definition 67899 : ;<87 =>

Staged module definition 9?8@AB CDB-7A E FG?H ,- . /0+−1 2 3 4 5

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Match pattern N I + J : J LM2 J 0

Match arm P I N ?HA. 5

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R J G, ! G9 P S Q J ' 4 $S Q J 7A? ' S Q

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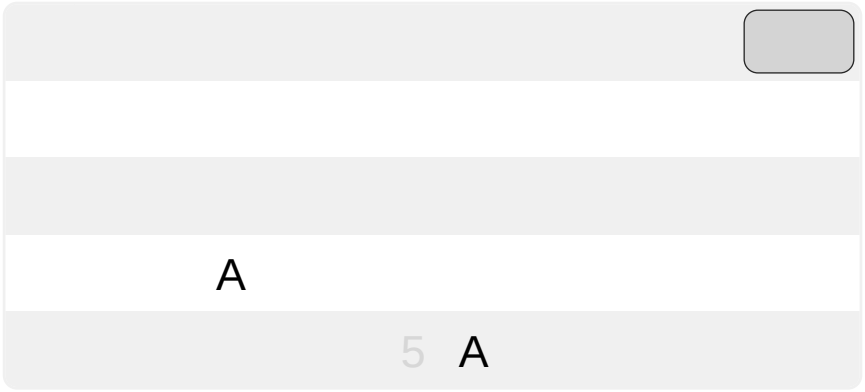

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Example

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Example

```
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Example

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Example

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